



EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	B.TECH	IIT Kharagpur	7.64 / 10
2022	CBSE(XII)	Kamla Devi Public School	88.6%
2020	CBSE(X)	Kamla Devi Public School	89.2%

INTERNSHIPS

Research Intern New Castle University Foreign Training Program <i>Objective: Map institutional enablers shaping Battery swapping ecosystems across regulatory, technological, industrial & policy dimensions</i> - Developed a structured battery-swapping database by using Python, compiling 2,500+ records across 15+ countries for institutional analysis - Analysed 1,200+ policies, patents & firm records spanning 10+ years and 12+ categories, to identify emerging key market and policy trends - Applied data mining, classification and comparative analysis across 3 institutional pillars, identifying 15+ key points for ecosystem development - Conducted sensitivity and robustness tests across 8 model specifications, deriving 5+ findings for battery-swapping policy & future investment	May 26 - Aug 26
AI/ML Intern Crescent Foundry Co. Pvt. Ltd. <i>Objective: Leverage data analytics, AI and process intelligence to optimize manufacturing performance, quality and cost outcomes</i> - Predicted cooling time using an AI model trained on 1,000+ records & 9 parameters, achieving a 22% cycle-time reduction through optimization - Integrated 3 IoT sensors across 7 monitoring stages into an AI curing system, automated control, anomaly detection and process monitoring - Identified 68% NVA lead time through Lean + VSM analysis, cutting the 62-day cycle by 30% through process and digital workflow improvement - Established vendor & paint-quality scorecards across 5+ quality, delivery & cost metrics, enabling supplier benchmarking & COPQ risk analysis	May 26 - Jul 26

PROJECTS

Portfolio Risk & Regime Analytics Self Project - Extracted multi-asset financial data spanning 10+ years, engineering 20+ technical and features for regime classification and portfolio analysis - Clustered market regimes using GMM and K-Means, identifying 3 distinct states from volatility, momentum and correlation based features - Quantified portfolio risk using 95% VaR, CVaR and 5+ stress scenarios, validating estimates through 1,000+ simulations and historical backtesting - Visualized portfolio performance via Streamlit dashboard, with 9% annualized returns, 12% volatility, 0.8 Sharpe, 1.1 Sortino & 15% drawdown	Aug 26 - Sep 26
PastPerfect Startup Project - Developed FastAPI + SQLAlchemy REST APIs for real-time ML predictions, handling 5,000+ concurrent requests with sub-50 ms average API latency - Built a real-time Demand Board (Inverse Marketplace) aggregating active buyer requests which ranks the top 4 campus categories within 50 ms - Architected Pillow + AWS S3 media pipelines achieving 60%+ image compression, with PhonePe integration for 100% cryptographic integrity - Trained Linear Regression + Bayesian models on 6,000+ datapoints across 50+ categories, achieving 85% price-forecasting model accuracy	May 25 - Sep 26
Financial Research & RAG Intelligence Self Project - Ingested 500+ SEC filings across 50+ companies, extracting financial statements, risk disclosures and commentary into 10,000+ data chunks - Benchmarked 3 retrieval strategies using Recall@5 and MRR, achieving 88% retrieval hit rate across 100+ diverse financial research queries - Evaluated RAG responses using Faithfulness and Answer Relevance, achieving 85% grounded responses with 18% measured hallucination rate - Deployed a FastAPI + Streamlit RAG platform with citation-backed answers, achieving 2 sec average query latency across 10+ financial metrics	July 26 - Aug 26
LLM Fine-Tuning for Autonomous Function Calling Self Project - Fine-tuned the Qwen2.5-Coder-1.5B-Instruct LLM with QLoRA and Unsloth on a Kaggle T4 GPU, efficient end-to-end function-calling training - Processed 60k samples of Salesforce's xLAM dataset, covering 3,600+ APIs spanning 21 domains with structured JSON tool-call supervision - Engineered 4-step Chain-of-Thought alignment strategy, generating reasoning traces to enforce parameter validation before JSON output - Designed an AST evaluation pipeline for tool selection and schema adherence, achieving 76% accuracy and a 91% valid JSON pass rate overall	May 25 - Jul 25

SKILLS AND EXPERTISE

AI/ML: Machine Learning Deep Learning NLP Transformers Federated Learning Computer Vision Anomaly Detection Time-Series
GenAI: Generative AI LLMs RAG LangChain LangGraph Reranking QLoRA AI Agents Hybrid Retrieval Embeddings Vector
ML Engineering & Software: PyTorch TensorFlow Scikit-Learn Pandas NumPy SQL MLOps APIs FastAPI Docker GitHub AWS
CS Fundamentals : Object Oriented Programming Operating Systems Database Management System LLD HLD Computer Networks

COURSEWORK INFORMATION

Computer Science: Programming and Data Structures* Software Engineering and Management Information Systems* Simulation*
Data and AI/ML: Data Analytics Statistical Learning* Distributed Analytics Advanced AI Paradigms Applied Reinforcement Learning
Mathematics and Optimization: Probability & Statistics Linear Algebra Calculus Operations Research I & II* Economics (*LAB included)

AWARDS AND ACHIEVEMENTS

Achieved a peak rating of 1527 on Codeforces with a best rank of 865 in Round 1025(Div. 2), and steadily improving through rated contests
Qualified NTSE Stage I, Madhya Pradesh , demonstrating strong analytical aptitude and consistent academic performance at the top state level

EXTRA CURRICULAR ACTIVITIES

Represented MS Hall in GC Product Management; Executive at MASK & Associate Member of Spring Fest and Shaurya , demonstrating leadership
Anchored OCAQ-24 , contributing to 130+ events and 80,000+ footfall; won an Open IIT Chess medal (Kharagpur) and published an original poem